

Qatar-1b

R-0035

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_260329 · created 2026-07-08T21:05:40+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461128.9219 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.022 +/- 0.056
Transit depth (Rp/Rs)^2	0.05 +/- 0.25 [%]
Orbital Inclination (inc)	88.71 +/- 2.92
Airmass coefficient 1 (a1)	0.36 +/- 0.21
Airmass coefficient 2 (a2)	0.78 +/- 0.48
Scatter in the residuals of the lightcurve fit is	52.5 %
Best Comparison Star	None
Optimal Aperture	1.87
Optimal Annulus	5.08
Transit Duration (day)	0.0714 +/- 0.0078

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.022 ± 0.056 vs 0.1463 (deviation 2.22σ)
Duration fit vs published	0.0714 d vs 0.0692 d (deviation 0.29σ)
Mid-time O–C	-7.9 min
Depth SNR	0.4
Residual scatter	52.5 %

Light curve

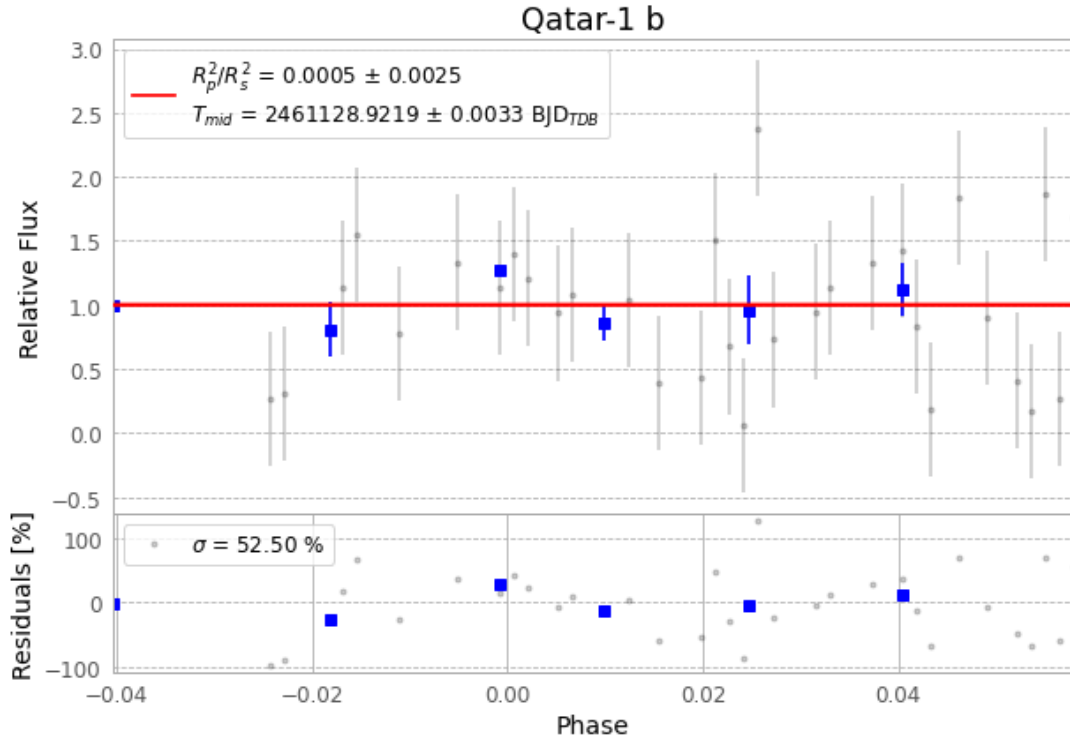


Figure 1 — FinalLightCurve_Qatar-1 b_2026-03-29.png (SHA-256 e52522e683214e57...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [291, 285], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 375462374f10e1ea...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

71 FITS frames (47 MB), Qatar-1260329083616.FITS → Qatar-1260329120615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-260329.log (073e260ae565...), FinalLightCurve_Qatar-1 b_2026-03-29.png (e52522e68321...), FinalLightCurve_Qatar-1 b_2026-03-29.csv (41a56b95b21e...)

Compute resources

Wall time 218.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.